

First-Principles Derivation of the Steiner Sentence Length Distribution

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Abstract

A *Steiner word* (Steiner circuit [1]) is a path segment of an odd Collatz orbit whose compressed mod-8 form matches $(7^*3)?(1 \mid 5)$: a sequence of Syracuse-map applications beginning at any letter in $\{1, 3, 5, 7\}$ and terminating at the terminal letter 1 or 5. A *Steiner sentence* is a string of $k \geq 1$ consecutive Steiner words in which all but the last word terminate at letter 1, and the final word terminates at letter 5; its *length* is the number of words k . We prove that the sentence length distribution is $P(\text{length} = k) = 3^{k-1}/4^k$ for all $k \geq 1$.

The proof models sentences as first-passage times of a Markov chain on $\{1, 3, 5, 7\} \pmod{8}$, whose transition matrix is computed row by row from elementary arithmetic. The key structural fact is an invariant $d_3 = d_7$ — equal weight on residue classes 3 and 7 in the non-terminated distribution — which holds unconditionally and forces the surviving mass to decay by exactly $3/4$ per word.

The proof requires no ergodic theory and no appeal to the Collatz conjecture. It rests on two arithmetic facts: that $v_2(3n+1)$ is a pointwise constant for $n \equiv 1, 3, 7 \pmod{8}$ (giving exact rational transition probabilities), and that $\gcd(3, 2^j) = 1$ (giving uniform output from $n \equiv 5 \pmod{8}$), proved in the companion paper [4]). All results have been machine-verified in Lean 4/Mathlib with no `sorry`.

Appendix B provides a complete empirical validation: 10^5 sentences sampled uniformly over $[1, 10^{15}]$, with χ^2 goodness-of-fit tests. The naive $(1/2)^k$ prediction is rejected with $\chi^2 \approx 10^6$; the formula $3^{k-1}/4^k$ is consistent ($p \approx 0.61$).

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1 Background and Notation

We work throughout with odd positive integers.

Definition 1.1 (Syracuse map). The *Syracuse map* on odd positive integers is

$$S(n) = \frac{3n + 1}{2^{v_2(3n+1)}}, \quad n \text{ an odd positive integer,}$$

where $v_2(m)$ denotes the 2-adic valuation of m .

Definition 1.2 (Steiner word). A *Steiner word* (or *Steiner circuit* [1, 2]) is a path segment of an odd Collatz orbit whose compressed mod-8 form matches the regular expression $(7^* 3)^?(1 \mid 5)$ [2]: a sequence of Syracuse-map applications whose successive input residues mod 8 spell a string over the alphabet $\{1, 3, 5, 7\}$ of that form, terminating when the input residue is 1 or 5 (the two *terminal letters*). The *entry letter* of a word is the residue modulo 8 of its first node; the *terminal letter* is 1 or 5.

Definition 1.3 (Steiner sentence). A *Steiner sentence* of length $k \geq 1$ is a string of k consecutive Steiner words in a Collatz orbit such that the first $k - 1$ words terminate at letter 1 and the final word terminates at letter 5.

Remark 1.4. A sentence of length k corresponds to a path of k edges in the overlay tree whose nodes are odd integers $\equiv 5 \pmod{8}$ and whose edges are individual Steiner words [2]: consecutive words in the sentence that terminate at letter 1 extend the current edge, while the word terminating at letter 5 completes it. This paper is self-contained; Paper 33 [2] (in preparation) develops the Steiner circuit decomposition, the regular-expression characterisation, and the overlay tree in full generality, but is not required here.

Remark 1.5 (Why Steiner sentences are the right scale). The Steiner sentence decomposition is chosen precisely so that class 5 (mod 8) is *always a sink and never a source* within a sentence. This has a crucial arithmetic consequence.

For $n \equiv 1, 3, 7 \pmod{8}$, the 2-adic valuation $v_2(3n + 1)$ is a *constant* determined entirely by $n \pmod{8}$: it equals 2, 1, and 1 respectively (Paper 66 [4], Theorem 2.1). The Syracuse image $S(n) \pmod{8}$ is therefore completely determined by $n \pmod{8}$ alone, with exact rational transition probabilities.

For $n \equiv 5 \pmod{8}$, by contrast, $v_2(3n+1) = 3 + v_2(3k+2)$ where $n = 8k+5$, and $v_2(3k+2)$ is an unbounded random variable whose expectation requires summing a geometric series (Paper 66). The transition probabilities out of class 5 are still exactly $1/4$ each, but they arise from a distribution over infinitely many valuation values, not a single fixed shift.

The Markov chain on entry letters tracks how the entry letter of each word determines the entry letter of the next, via the transition matrix P . Letter 5 is the unique terminal letter that ends a sentence; letter 1 is the other terminal letter, but it begins the next word rather than ending the sentence. Within a sentence the surviving chain operates on entry letters in $\{1, 3, 7\}$: a word with entry letter 5 would have terminated the *previous* sentence, so letter 5 never acts as an entry letter for a non-initial word inside a sentence. For entry letters 1, 3, and 7, the 2-adic valuation $v_2(3n+1)$ is a *constant* determined by the entry letter alone, so every transition probability is an exact rational. The variable-valuation complexity of letter 5 is absorbed at the sentence boundary and plays no role in the counting argument. This is why the sentence length distribution $3^{k-1}/4^k$ admits a clean closed-form proof: at the scale of Steiner sentences (strings of words), the arithmetic is exact.

Paper 65 [3] was the first attempt at this subject. It observed empirically that $P(\text{sentence length} = k) = \frac{1}{3}(3/4)^k$, equivalently $3^{k-1}/4^k$, and stated the result as a conjecture. The present paper supersedes Paper 65 by providing a complete proof, and incorporates its experimental record in Appendix B.

2 The Syracuse Transition Matrix mod 8

We model the sentence as a Markov chain on the *entry letter* of each successive Steiner word — the residue modulo 8 of the word's first node, a letter in the alphabet $\{1, 3, 5, 7\}$. Each word terminates at a terminal letter (1 or 5); this terminal letter is the entry letter of the *next* word (since the terminal node of one word is the starting node of the next). A sentence ends when a word terminates at letter 5. The transition probability $P[a][b]$ is therefore the natural density of odd integers $n \equiv a \pmod{8}$ for which $S(n) \equiv b \pmod{8}$:

$$P[a][b] = \lim_{N \rightarrow \infty} \frac{1}{N} \#\{n \leq N : n \equiv a \pmod{8}, S(n) \equiv b \pmod{8}\}.$$

All results in this paper are unconditional: they require no assumption beyond this standard density definition and do not depend on the Collatz conjecture.

Theorem 2.1 (Syracuse transition matrix mod 8). *The transition matrix on $\{1, 3, 5, 7\} \pmod{8}$ is:*

$$P = \begin{array}{c|cccc} & 1 & 3 & 5 & 7 \\ \hline 1 & 1/4 & 1/4 & 1/4 & 1/4 \\ 3 & 1/2 & 0 & 1/2 & 0 \\ 5 & 1/4 & 1/4 & 1/4 & 1/4 \\ 7 & 0 & 1/2 & 0 & 1/2 \end{array}$$

(rows = source letter, columns = target letter).

Proof. We prove each row by direct arithmetic.

Row 1 (uniform). Write $n = 8k + 1$. Then $3n + 1 = 24k + 4 = 4(6k + 1)$. Since $6k + 1$ is always odd, $v_2(3n + 1) = 2$ exactly and $S(n) = 6k + 1$. Now $6k + 1 \pmod{8}$: as k runs through $0, 1, 2, 3 \pmod{4}$, we get $6k + 1 \equiv 1, 7, 5, 3 \pmod{8}$ respectively. Each residue in $\{1, 3, 5, 7\}$ appears with equal density, so $P[1][b] = 1/4$ for all $b \in \{1, 3, 5, 7\}$.

Row 5 (uniform). Write $n = 8k + 5$. Then $3n + 1 = 8(3k + 2)$, so $v_2(3n + 1) = 3 + v_2(3k + 2)$. Write $3k + 2 = 2^m q$ with q odd; then $S(n) = q$. We need $q \pmod{8}$ to be uniform on $\{1, 3, 5, 7\}$.

Since $\gcd(3, 2^j) = 1$ for all $j \geq 1$, the map $k \mapsto 3k + 2$ is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$. As k ranges over any 2^{j+3} consecutive integers, the value $3k + 2$ hits each residue class mod 2^{j+3} exactly 2^j times. Consequently the value $S(8k + 5) \pmod{8}$ is equidistributed over $\{1, 3, 5, 7\}$ *in natural density*: the arithmetic core is that, as k ranges over $\{0, 1, \dots, 2^N - 1\}$, exactly 2^{N-j} values satisfy $2^j \mid (3k + 2)$ (Paper 66 [4], `uniformity_count`; formalised here as `row5_uniform_dvd`), from which each target residue $b \in \{1, 3, 5, 7\}$ receives limiting density $1/4$. Thus $P[5][b] = 1/4$ for all $b \in \{1, 3, 5, 7\}$.

Remark 2.2 (Row 5 is a density, not an exact finite count). Unlike rows 1, 3 and 7 — whose per-residue counts over $\{0, \dots, 2^N - 1\}$ are *exactly* 2^{N-2} or 2^{N-1} (formalised as `count_row1`, `count_row3_*`, `count_row7_*`) — row 5 holds only as a natural-density limit. The finite count $\#\{k < 2^N : S(8k + 5) \equiv b \pmod{8}\}$ is *not* equal to 2^{N-2} in general: at $N = 6$, $b = 1$ the count is 17, not $16 = 2^{6-2}$. This is machine-verified in the companion Lean development by `row5_count_not_exact` (via `native_decide`). The extra, variable number of halvings $v_2(3n + 1) = 3 + v_2(3k + 2)$ from class 5 is exactly what prevents an exact finite count while still forcing uniform limiting density.

Row 3 (restricted to $\{1, 5\}$). Write $n = 8k + 3$. Then $3n + 1 = 24k + 10 = 2(12k + 5)$. Since $12k$ is even, $12k + 5$ is always odd, so $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 5$. Now $12k + 5 \pmod{8}$: $12k \equiv 4k \pmod{8}$, so $12k + 5 \equiv 4k + 5 \pmod{8}$. For k odd: $4k \equiv 4 \pmod{8}$, giving $12k + 5 \equiv 1 \pmod{8}$. For k even: $4k \equiv 0 \pmod{8}$, giving $12k + 5 \equiv 5 \pmod{8}$. Since k is odd and even with equal density, $P[3][1] = P[3][5] = 1/2$ and $P[3][3] = P[3][7] = 0$.

Row 7 (restricted to $\{3, 7\}$). Write $n = 8k + 7$. Then $3n + 1 = 24k + 22 = 2(12k + 11)$. Since $12k + 11$ is always odd, $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 11$. Now $12k + 11 \pmod{8}$: $12k + 11 \equiv 4k + 3 \pmod{8}$. For k odd: $4k \equiv 4$, giving $4k + 3 \equiv 7 \pmod{8}$. For k even: $4k \equiv 0$, giving $4k + 3 \equiv 3 \pmod{8}$. Therefore $P[7][3] = P[7][7] = 1/2$ and $P[7][1] = P[7][5] = 0$. $\square$

Remark 2.3 (Forbidden transitions). The zero entries in P — transitions $3 \rightarrow 3$, $3 \rightarrow 7$, $7 \rightarrow 1$, $7 \rightarrow 5$ — are *structurally forbidden* for all n , not just on average. Classes 3 and 7 each map into a strict subset of $\{1, 3, 5, 7\}$. This structural restriction is the engine of the proof.

Remark 2.4 (Machine verification). All results in this theorem have been formally verified in Lean 4/Mathlib with no `sorry` or `admit` and no axioms beyond `propext`, `Classical.choice`, and `Quot.sound`. The key theorems are `S_class1/3/5/7` (arithmetic identities), `S_class1_mod8` (row 1 uniformity by direct mod-4 computation), `count_row1/3/7` (exact finite counts for rows 1, 3, 7), and `row5_count_not_exact` (counterexample for row 5 via `native_decide`: at $N = 6$, $b \equiv 1$ the count is 17, not 16). The Lean source is in `Paper67.lean` in the paper directory.

Remark 2.5 (The surviving chain has exact arithmetic). Row 5 of P is never used in the proof of Theorem 5.1. Within a sentence, class 5 is a sink: once the chain reaches 5 the sentence ends, and class 5 never acts as a source for subsequent words. The surviving distribution (Definition 3.2) propagates mass only through rows 1, 3, and 7 of P .

For these three rows, all transition probabilities are *exact rational constants* that follow directly from $n \bmod 8$ alone: $v_2(3n+1) = 2$ always for class 1, and $v_2(3n+1) = 1$ always for classes 3 and 7. No geometric series, no expectation, no appeal to the distribution of $v_2(3n+1)$ for class 5 is ever needed. The variable-valuation behaviour of class 5 — which requires an infinite geometric series to characterise (Paper 66 [4]) — is invisible to the surviving chain. This exactness is what makes the sentence decomposition the right scale for counting, and what makes the proof of Theorem 5.1 purely algebraic.

3 The Sentence as a First-Passage Problem

A sentence is the first-passage time of the Markov chain (P, π_0) to state 5, where π_0 is the sentence-start distribution.

Lemma 3.1 (Sentence-start distribution). *The sentence-start distribution is $\pi_0 = (1/4, 1/4, 1/4, 1/4)$ — uniform on $\{1, 3, 5, 7\}$.*

Proof. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$. By Theorem 2.1, row 5 of P is uniform, so $b_0 \bmod 8$ is uniform on $\{1, 3, 5, 7\}$. $\square$

The sentence terminates at the first word where the chain visits state 5. The sentence length k is this first-passage time.

Definition 3.2 (Surviving distribution). Let $d_s^{(k)}$ denote the probability that the chain is at state s after k words without having visited state 5. Formally, $d_s^{(0)} = \pi_0(s)$ and for $k \geq 0$:

$$d_s^{(k+1)} = \sum_{t \neq 5} d_t^{(k)} P[t][s], \quad s \neq 5.$$

The *surviving mass* is $M_k = \sum_{s \neq 5} d_s^{(k)}$.

The sentence length distribution follows from the surviving mass: M_{k-1} is the probability of surviving past word $k-1$, and M_k is the probability of surviving past word k , so

$$P(\text{sentence length} = k) = M_{k-1} - M_k.$$

Expanding: $M_{k-1} - M_k = \sum_{s \neq 5} d_s^{(k-1)} - \sum_{s \neq 5} d_s^{(k)} = \sum_{t \neq 5} d_t^{(k-1)} \left(1 - \sum_{s \neq 5} P[t][s]\right) = \sum_{t \neq 5} d_t^{(k-1)} P[t][5].$

4 The Invariant $d_3 = d_7$

The key structural fact is a symmetry between classes 3 and 7 in the surviving distribution.

Lemma 4.1 (Invariant). $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Proof. By Definition 3.2, the surviving distribution sums only over sources $t \neq 5$; class 5 contributes nothing because once the chain reaches 5 the sentence terminates (the mass is absorbed, not propagated). The transition update for states 3 and 7 is therefore:

$$\begin{aligned} d_3^{(k+1)} &= d_1^{(k)} \cdot P[1][3] + d_3^{(k)} \cdot P[3][3] + d_7^{(k)} \cdot P[7][3] \\ &= \frac{1}{4} d_1^{(k)} + 0 \cdot d_3^{(k)} + \frac{1}{2} d_7^{(k)}, \\ d_7^{(k+1)} &= d_1^{(k)} \cdot P[1][7] + d_3^{(k)} \cdot P[3][7] + d_7^{(k)} \cdot P[7][7] \\ &= \frac{1}{4} d_1^{(k)} + 0 \cdot d_3^{(k)} + \frac{1}{2} d_7^{(k)}. \end{aligned}$$

The two expressions are identical: $d_3^{(k+1)} = d_7^{(k+1)}$ for any $d_1^{(k)}, d_3^{(k)}, d_7^{(k)}$. By induction, since $d_3^{(0)} = d_7^{(0)} = 1/4$ (Lemma 3.1), $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$. $\square$

Remark 4.2 (Structural source of the invariant). The invariant holds because columns 3 and 7 of P , restricted to rows $\{1, 3, 7\}$, are *identical*:

$$P[\cdot][3]|_{\{1,3,7\}} = P[\cdot][7]|_{\{1,3,7\}} = \left(\frac{1}{4}, 0, \frac{1}{2}\right).$$

This column symmetry is a direct consequence of Theorem 2.1:

- *From class 1*: uniform output gives equal weight $1/4$ to both 3 and 7.
- *From class 3*: forbidden to reach 3 or 7 (both have weight 0).
- *From class 7*: reaches 3 and 7 with equal probability $1/2$ each.

The invariant is unconditional — it holds for *any* value of d_1 , not just at the fixed point of the chain.

5 Main Theorem

Theorem 5.1 (Sentence length distribution). *For any $k \geq 1$:*

$$P(\text{sentence length} = k) = \frac{3^{k-1}}{4^k}.$$

Equivalently, $P(\text{sentence length} = k) = \frac{1}{3} \left(\frac{3}{4}\right)^k$.

Proof. We use the splitting: $P(\text{sentence length} = 1) = \pi_0(5) = 1/4$ (sentences that start at $b_0 \equiv 5$ terminate immediately). For $k \geq 2$, the sentence does not terminate at word 1 (probability $3/4$) and then terminates at word $k-1$ from the conditional start, i.e. $P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1}$, where g_j is the first-passage probability to 5 starting from the conditional distribution $\pi_0(\cdot | \cdot \neq 5) = (1/3, 1/3, 1/3)$ uniform on $\{1, 3, 7\}$.

It suffices to show $g_j = (3/4)^{j-1}/4$ for all $j \geq 1$, which follows if the surviving mass M_j starting from uniform $\{1, 3, 7\}$ satisfies $M_j = (3/4)^j$.

Step 1: $d_3 = d_7$ is preserved. The conditional start has $d_3^{(0)} = d_7^{(0)} = 1/3$, so Lemma 4.1 applies: $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Step 2: Surviving mass decays by exactly $3/4$. From Definition 3.2, $M_{k+1} = \sum_{s \neq 5} d_s^{(k+1)} = \sum_{s \neq 5} \sum_{t \neq 5} d_t^{(k)} P[t][s] = \sum_{t \neq 5} d_t^{(k)} \sum_{s \neq 5} P[t][s] = \sum_{t \neq 5} d_t^{(k)} (1 - P[t][5])$, giving:

$$M_{k+1} = d_1^{(k)} (1 - P[1][5]) + d_3^{(k)} (1 - P[3][5]) + d_7^{(k)} (1 - P[7][5]).$$

Substituting $P[1][5] = 1/4$, $P[3][5] = 1/2$, $P[7][5] = 0$:

$$M_{k+1} = \frac{3}{4} d_1^{(k)} + \frac{1}{2} d_3^{(k)} + d_7^{(k)}.$$

We claim $M_{k+1} = \frac{3}{4} M_k = \frac{3}{4} (d_1^{(k)} + d_3^{(k)} + d_7^{(k)})$. This requires:

$$\frac{3}{4} d_1^{(k)} + \frac{1}{2} d_3^{(k)} + d_7^{(k)} = \frac{3}{4} d_1^{(k)} + \frac{3}{4} d_3^{(k)} + \frac{3}{4} d_7^{(k)},$$

which simplifies to $\frac{1}{2} d_3^{(k)} + d_7^{(k)} = \frac{3}{4} (d_3^{(k)} + d_7^{(k)})$, i.e. $d_7^{(k)} = d_3^{(k)}$. This is exactly Lemma 4.1.

Step 3: Initial condition. At $k = 0$, $M_0 = 1$ (the full conditional mass on $\{1, 3, 7\}$). By induction, $M_k = (3/4)^k$ for all $k \geq 0$.

Conclusion. $g_j = M_{j-1} - M_j = (3/4)^{j-1} - (3/4)^j = (3/4)^{j-1} (1/4) = 3^{j-1}/4^j$. Therefore:

$$P(\text{length} = 1) = \frac{1}{4} = \frac{3^0}{4^1},$$

$$P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1} = \frac{3}{4} \cdot \frac{3^{k-2}}{4^{k-1}} = \frac{3^{k-1}}{4^k}, \quad k \geq 2.$$

Both cases give $3^{k-1}/4^k$, completing the proof. □

Corollary 5.2 (Normalisation). $\sum_{k=1}^{\infty} P(\text{sentence length} = k) = 1$.

Proof. $\sum_{k=1}^{\infty} \frac{3^{k-1}}{4^k} = \frac{1}{4} \sum_{k=1}^{\infty} \left(\frac{3}{4}\right)^{k-1} = \frac{1}{4} \cdot \frac{1}{1-3/4} = 1$. □

Remark 5.3 (Machine verification of the Markov chain algebra). The Markov chain results — Lemma 4.1 (`invariant_d3_d7`), $M_k = (3/4)^k$ (`Mmass_eq`), $g_j = 3^{j-1}/4^j$ (`g_eq`), $P(\text{length} = k) = 3^{k-1}/4^k$ (`Plen_eq`), and the normalisation identity (`normalization`) — have all been formally verified in Lean 4/Mathlib with no `sorry`, no `admit`, and no axioms beyond the standard Lean kernel. The verification operates on exact rational arithmetic throughout; no real-analysis machinery is required. The Lean source is in `Paper67.lean` in the paper directory.

6 Discussion

6.1 The role of class 5

The formula $3^{k-1}/4^k$ is the geometric distribution with success probability $1/4$, as if at each word the chain draws uniformly from $\{1, 3, 5, 7\}$ and terminates on drawing 5. This

appearance is striking because class 5 plays three distinct and mutually reinforcing roles in the sentence structure.

Class 5 as the termination state. A sentence terminates exactly when a word’s terminal letter is 5. Since letter 5 is the unique absorbing boundary of the chain (once the chain’s terminal letter is 5, the sentence ends), letter 5 is structurally absent as an entry letter for all intermediate words. Empirically, letter 5 accounts for only $\approx 6.2\%$ of within-sentence word entries (Appendix B, Table 2) — one entry per sentence, uniformly distributed across sentence lengths — rather than the 25% it would contribute if words were unrestricted.

Class 5 as the uniform source. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$. By row 5 of the transition matrix (Theorem 2.1), letter 5 maps uniformly over $\{1, 3, 5, 7\}$: any of the four letters is equally likely as the entry letter of the first word. In particular, exactly $1/4$ of sentences have $b_0 \equiv 5$, meaning length $k = 1$. This is the correct $P(k = 1) = 1/4$, not the naive $1/2$. Class 5 is the unique class for which $v_2(3n + 1) \geq 3$ always (proved in Paper 66 [4]), which forces the extra halvings that spread the output uniformly.

Class 5 as the structural asymmetry. The transition $7 \rightarrow 5$ is structurally forbidden: $P[7][5] = 0$. Class 5 is therefore unreachable *directly* in one word from class 7, while it is reachable from classes 1 (with probability $1/4$) and 3 (with probability $1/2$). This asymmetry — class 5 reachable from some classes but not others — is precisely what forces the invariant $d_3 = d_7$: nodes in class 7 can only reach class 5 via a detour through class 3 or class 7, never directly. The proof in Section 4 shows that this forced detour maintains equal weight on classes 3 and 7 throughout the chain, and Section 5 shows that this equal weight is the algebraic condition that makes the effective termination probability exactly $1/4$ at every step, regardless of the current value of d_1 .

6.2 Connection to Paper 66

Paper 66 [4] proves that $E[v_2(3n+1) \mid n \equiv r \pmod{8}]$ equals exactly 2, 1, 4, 1 for $r = 1, 3, 5, 7$ respectively. The two papers are complementary: Paper 66 establishes what happens *at* the class 5 boundary (a geometrically distributed valuation requiring an infinite series for its expectation), while the present paper shows that by choosing Steiner sentences as the unit of analysis, the class 5 boundary need never be crossed from the inside — it only needs to be *entered*.

More precisely, the $3/4$ decay rate follows from three facts, each drawing on the same underlying arithmetic:

- The $1/4$ probability of immediate termination ($b_0 \equiv 5$) uses row 5 of P exactly once — at the sentence start. Class 5 is the unique class for which $v_2(3n + 1) \geq 3$ always (Theorem 2.1(3) of Paper 66 [4]), forcing the extra halvings that make its output uniform. This is the only point where the geometric-series character of class 5 is relevant.
- The $1/2$ termination probability from class 3 comes from the mod-16 split of Theorem 2.1: $v_2(3n + 1) = 1$ exactly for all $n \equiv 3 \pmod{8}$, so $S(n) \bmod 8$ is determined by $n \bmod 16$ alone, with no random valuation.

- The invariant $d_3 = d_7$ — and hence the $3/4$ decay — involves classes 1, 3, and 7 only, all of which have constant valuations. The proof is purely algebraic: no expectation, no series.

Paper 66 handles the hard case (class 5, variable valuation). Paper 67 sidesteps it by construction.

6.3 Why $3^{k-1}/4^k$ rather than $(1/2)^k$

The naive prediction $(1/2)^k$ arises from treating each step as an independent trial that terminates whenever the current word is in class 3 (with probability $1/2$, since $P[3][5] = 1/2$) or class 7 (also $1/2$ by this model). If the within-sentence distribution were uniform over $\{1, 3, 5, 7\}$, the effective per-step termination probability would be:

$$\frac{1}{4} \cdot 0 + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2},$$

where the terms correspond to classes 1 (termination probability 0 in this model), 3 ($1/2$), 5 (terminal, probability 1), and 7 ($1/2$). This model is wrong in two ways. First, class 1 *can* reach class 5 directly: $P[1][5] = 1/4$, so the naive assignment of 0 to class 1 understates termination from class 1. Second and more importantly, class 7 *cannot* reach class 5 *directly*: $P[7][5] = 0$, so the naive assignment of $1/2$ to class 7 overstates it. The net effect of these two corrections is a reduction in the effective termination rate from $1/2$ to $1/4$. The structural reason is the forbidden transition $7 \rightarrow 5$: a word with entry letter 7 must first transit to letter 3 or 7 (remaining in the growth letters), where it may then terminate at letter 5 on a subsequent word. The invariant $d_3 = d_7$ (Lemma 4.1) shows that these two errors cancel exactly and yield an effective rate of precisely $1/4$, independent of d_1 .

Acknowledgements

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A The Steiner Language Hierarchy

The objects studied in this paper — sentences, words, letters — belong to a hierarchy of structures that together form a language-theoretic description of Collatz orbits. This appendix defines the hierarchy from the bottom up and connects it to the Collatz conjecture. The terminology is specific to this body of work; the foundational combinatorial infrastructure is developed in Paper 33 [2].

The Steiner Alphabet and Letters

The *Steiner alphabet* is the set $\{1, 3, 5, 7\}$ of odd residues modulo 8. Each element of the alphabet is a *Steiner letter*, encoding the step type of a single application of the Syracuse map:

Letter	Input class	$v_2(3n + 1)$
1	$n \equiv 1 \pmod{8}$	2 (OEE step)
3	$n \equiv 3 \pmod{8}$	1 (OE step)
5	$n \equiv 5 \pmod{8}$	≥ 3 (OEEE ⁺ step)
7	$n \equiv 7 \pmod{8}$	1 (OE step)

Letters 1, 3, and 7 have *constant* 2-adic valuations determined entirely by the residue class. Letter 5 has a *variable* valuation $v_2(3n + 1) = 3 + v_2(3k + 2)$ (where $n = 8k + 5$) that depends on k and is unbounded; its statistical behaviour requires an infinite geometric series and is studied in Paper 66 [4]. The two *terminal letters* are 1 and 5: these are the only letters at which a Steiner word can end.

Steiner Words

A *Steiner word* (or *Steiner circuit* [1, 2]) is a path segment of an odd Collatz orbit whose sequence of input residues mod 8 matches the regular expression

$$(7^* 3)?(1 \mid 5).$$

That is: zero or more letters 7, optionally followed by one letter 3, followed by exactly one terminal letter (1 or 5). Every Steiner word terminates at a terminal letter; words terminating at letter 1 are *OEE-terminal* and words terminating at letter 5 are *OEEE⁺-terminal*. The *entry letter* of a word is its first letter (the residue mod 8 of its starting node); the *terminal letter* is its last letter (1 or 5).

Every Collatz orbit decomposes uniquely into a sequence of Steiner words (Proposition 2.1 of Paper 33 [2]).

Remark A.1 (Word vs. circuit). Steiner [1] introduced the term *circuit* for these path segments, and that term remains standard in the Collatz literature. We prefer *word* here because the present framework embeds circuits into a richer linguistic hierarchy: words compose into sentences, sentences into volumes, and volumes into libraries. Within this hierarchy, *word* is the natural term — it carries the right connotations of a discrete unit that can be concatenated and counted, and it aligns with standard formal-language usage where words are strings over an alphabet. The term *circuit* is retained as a synonym (and in the compound *Steiner circuit*) where it aids clarity or connection to prior work.

Steiner Sentences

A *Steiner sentence* of length $k \geq 1$ is a string of k consecutive Steiner words in a Collatz orbit such that the first $k - 1$ words are OEE-terminal (terminal letter 1) and the k -th word is OEEE⁺-terminal (terminal letter 5).

Equivalently, a sentence is a path of k edges in the $5 \pmod{8}$ *overlay tree*: the directed graph whose nodes are odd integers $\equiv 5 \pmod{8}$ and whose edges are individual Steiner words [2]. The sentence length distribution $P(\text{length} = k) = 3^{k-1}/4^k$ is the main result of this paper (Theorem 5.1).

Steiner Volumes

A *Steiner Volume* is defined recursively as an ordered pair

$$V = (\sigma, V'),$$

where σ is a Steiner sentence and V' is a Steiner Volume. Every Volume has a *root node* $r \equiv 5 \pmod{8}$. The sentence σ is an OEEE-free path beginning at a node l and ending at r : no word in σ has entry letter 5, so no node that serves as the starting node of any word in σ is $\equiv 5 \pmod{8}$. (The terminal node $r \equiv 5 \pmod{8}$ is the endpoint of σ , not the entry of any word within it.) The sub-Volume V' is rooted at the node r' satisfying

$$S(r') = r, \quad r' \equiv 5 \pmod{8},$$

i.e. r' is the unique $5 \pmod{8}$ predecessor of r under S ($v_2(3r' + 1) \geq 3$). Thus the recursive structure of a Volume follows the *inverse* of the Syracuse map: the root r of V is the forward image of the root r' of V' .

Postulate A.2 (Left-sentence origin). For every Steiner Volume with root $r \equiv 5 \pmod{8}$, the OEEE-free sentence σ begins at a node $l \equiv 0 \pmod{3}$.

This postulate is supported by extensive computation but has not been proved.

Remark A.3. This property was first isolated in early unpublished structural drafts (2025) and is adopted here axiomatically as an operational postulate native to this paper.

Note that $r \equiv 5 \pmod{8}$ and $r \equiv 0 \pmod{3}$ are compatible ($r \equiv 21 \pmod{24}$); every third Volume root satisfies both congruences simultaneously. For all other roots the postulate asserts that σ begins at a distinct node $l \equiv 0 \pmod{3}$ whose forward path under S ends at r .

Steiner Libraries

A *Steiner Library* is a collection of Steiner Volumes sharing a common root node b . The known library has root $b = 1$ and contains the Volumes rooted at

$$5, 21, 85, 341, \dots \quad (a_0 = 5, a_{n+1} = 4a_n + 1),$$

each of which maps to 1 under a single application of S ($S(5) = 1, S(21) = 1, S(85) = 1, \dots$).

The Steiner Library is the natural object at which to state the Collatz conjecture:

Conjecture A.4 (Library Equivalence). The global Collatz conjecture is equivalent to the structural assertion that there exists exactly one Steiner Library — rooted at $b = 1$ — to which every odd positive integer belongs.

The two directions of Conjecture A.4 correspond to the only two ways the conjecture can fail:

- A counterexample forming a *cycle* would give rise to a second Library whose Volumes circulate among themselves without reaching 1.
- A counterexample *escaping to infinity* would give rise to a Library with no finite root.

The sentence length distribution proved in this paper (Theorem 5.1) is a property of *all* Steiner Libraries, unconditionally: the proof rests on arithmetic facts about $v_2(3n+1)$ that hold for every odd positive integer, regardless of which Library it belongs to or whether the Collatz conjecture is true.

B Empirical Validation

This appendix records the complete numerical experiment establishing the sentence length distribution. The experiment was first reported in Paper 65 [3] and is reproduced here in full as the empirical record accompanying this paper’s proof.

Sampling Protocol

1. Draw t uniformly at random from $[1, T_{\max}]$ with $T_{\max} = 10^{15}$.
2. Set $n = 8t + 5$ (so $n \equiv 5 \pmod{8}$).
3. Compute $b_0 = S(n)$, the first Steiner word of the new sentence.
4. Follow consecutive Steiner circuits from b_0 , counting words, until the first output $b \equiv 5 \pmod{8}$. Record the word count as k .

$N = 10^5$ independent sentences were collected with seed 42. Independence between samples is ensured by the random draw of t ; no orbit tails are shared.

Sentence-start Distribution

By Theorem 2.1, row 5 of P is uniform, so $b_0 = S(n) \bmod 8$ should be uniform on $\{1, 3, 5, 7\}$.

$b_0 \bmod 8$	Count	Observed fraction	Expected (uniform)
1	25181	0.25181	0.25000
3	25023	0.25023	0.25000
5	24852	0.24852	0.25000
7	24944	0.24944	0.25000

Table 1: Sentence-start distribution: $b_0 = S(n)$, $n \equiv 5 \pmod{8}$, $N = 10^5$ samples. The distribution is uniform on $\{1, 3, 5, 7\}$ as predicted by Theorem 2.1.

Within-sentence Steiner Word Entry Distribution

The entry letter of each Steiner word within a sentence determines its termination behaviour. Words with entry letter 5 always terminate the sentence, so they appear only as the final word — their within-sentence frequency is structurally suppressed to approximately $1/(k+1)$ for a sentence of length k , averaging to $\approx 1/16$ empirically.

Entry mod8	Count	Fraction	Can reach class 5 directly?	Term. prob.
1	125450	0.31333	yes ($P_{15} = 1/4$)	$1/4$
3	125119	0.31250	yes ($P_{35} = 1/2$)	$1/2$
5	24852	0.06207	— (terminal only)	1
7	124955	0.31209	no ($P_{75} = 0$)	0
Total	400376			

Table 2: Within-sentence Steiner word entry distribution across all 10^5 sentences (400376 total words). Class 5 appears only as the terminal word, giving frequency $\approx 6.2\%$ rather than the uniform 25%. The last column gives the probability of a word terminating at letter 5 given its entry letter ($P_{15} = 1/4$, $P_{35} = 1/2$, $P_{75} = 0$) — these are the exact entries of column 5 of the transition matrix P (Theorem 2.1). The invariant $d_3 = d_7$ (Lemma 4.1) makes the effective per-word termination probability exactly $1/4$ (Theorem 5.1).

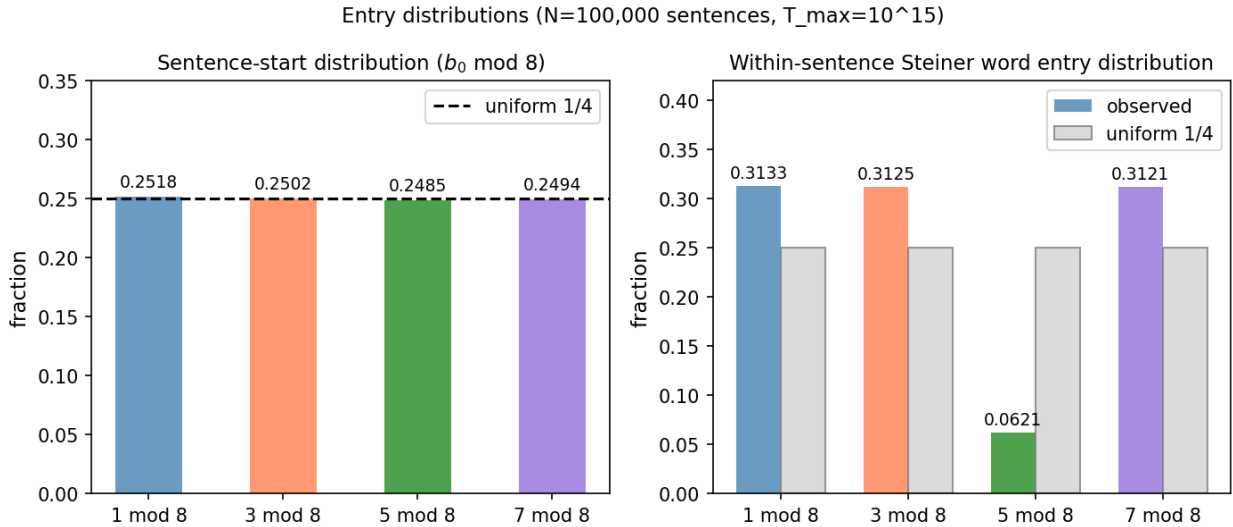


Figure 1: Left: sentence-start distribution ($b_0 \bmod 8$) against the uniform $1/4$ baseline — confirmed uniform (Table 1). Right: within-sentence Steiner word entry distribution against the uniform $1/4$ baseline. Class 5 is structurally suppressed to $\approx 6.2\%$ (it can only appear as the terminal word); classes 1, 3, 7 each carry $\approx 31\%$.

k	Observed fraction	Naive $(1/2)^k$	Theory $3^{k-1}/4^k$
1	0.25180	0.50000	0.25000
2	0.18683	0.25000	0.18750
3	0.13931	0.12500	0.14063
4	0.10574	0.06250	0.10547
5	0.07844	0.03125	0.07910
6	0.05958	0.01563	0.05933
7	0.04490	0.00781	0.04449
8	0.03342	0.00391	0.03337
9	0.02490	0.00195	0.02503
10	0.01866	0.00098	0.01877

Table 3: Empirical Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$) vs. the naive $(1/2)^k$ prediction and the theoretical $3^{k-1}/4^k$ (Theorem 5.1). The naive prediction overestimates $P(k = 1)$ by a factor of 2 and decays far too fast for $k \geq 3$.

Sentence Length Distribution

Statistical Tests

A χ^2 goodness-of-fit test on 15 length bins plus a tail bin, using $N = 10^5$ sentences:

Model	χ^2	p -value	Verdict
Naive $(1/2)^k$	1,022,029	≈ 0	Rejected
Theory $3^{k-1}/4^k$	12.9	0.61	Consistent

The naive model is rejected with overwhelming confidence. The theoretical formula $3^{k-1}/4^k$ is consistent with the data at all tested lengths.

References

- [1] R. P. Steiner, A theorem on the Syracuse problem, *Proceedings of the 7th Manitoba Conference on Numerical Mathematics and Computing*, Winnipeg, 1977, pp. 553–559.
- [2] J. Seymour, *A Regular Expression Language for the Collatz Graph*, in preparation, 2026.
- [3] J. Seymour, *The $\frac{1}{3}(\frac{3}{4})^k$ Distribution of Steiner Sentence Lengths* (superseded by the present paper), 2026.
- [4] J. Seymour, *Exact 2-adic Valuations of $3n+1$ and the Geometric Mean Contraction Rate of the Syracuse Map* (companion paper, submitted simultaneously), 2026.
- [5] J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, Providence, RI, 2010.

Steiner circuit sentence length distribution ($N = 100,000$ sentences, $T_{\max} = 10^{15}$)

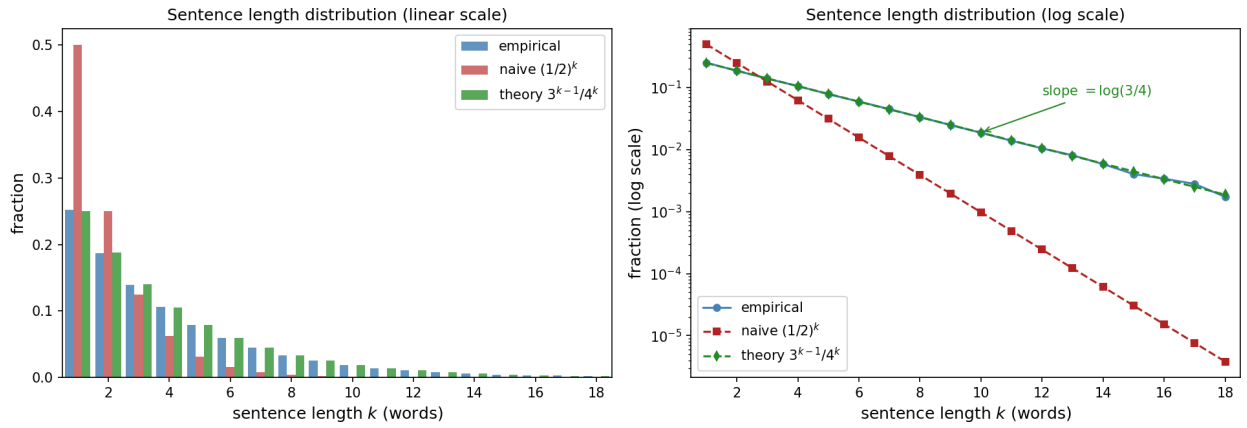


Figure 2: Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$): empirical (blue bars / circles) vs. naive $(1/2)^k$ (red) and theoretical $3^{k-1}/4^k$ (green), on linear scale (left) and log scale (right). On the log scale, the empirical slope matches $\log(3/4)$ precisely — not $\log(1/2)$. The naive prediction is dramatically refuted at $k = 1$ (predicts 50%; observed 25%) and for $k \geq 3$ (predicts too-fast decay).